



Internship Report On:

“ CYBERSECURITY “

At

Shadowfox

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Submitted by:

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TASK LEVEL BEGINNER

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Introduction

Cybersecurity helps to find and fix security problems in websites. If security problems are not fixed, hackers may misuse them.

During my internship at ShadowFax, I tested the security of this website:
<http://testasp.vulnweb.com>

This website is made for learning and practice, so it is safe to test legally.

Tasks Covered

1. Port Scanning – Finding all open ports on the website
2. Directory Brute Forcing – Discovering hidden folders on the website
3. Network Traffic Sniffing – Capturing login credentials using Wireshark

Tools Used

- Kali Linux Operating System
- Nmap – for port scanning
- Dirbuster – for directory brute forcing
- Wireshark – for capturing network traffic
- Firefox Browser

Task 1 – Port Scanning

Find open ports on the website: **<http://testasp.vulnweb.com>**

Attack Type : Port Scanning

Why Is This Important?

- Open ports can be used by attackers to enter a system. If a port is not secure, hackers may misuse it.
- In this website, port 80 was open. Port 80 uses HTTP, so the data is not encrypted. This means attackers may see the data sent between the user and the website.

Step 1 – Find IP Address

First, the IP address of the website was found using the ping command.

Step 2 – Scan Ports Using Nmap

After finding the IP address, Nmap was used to scan the ports.

Scan Details

- Website: <http://testasp.vulnweb.com>
- IP Address:192.168.1.1
- Command:

```
nmap -sV 192.168.1.1
```

```
Session Actions Edit View Help
root@kali: ~
(kali@kali)-[~]
$ sudo -i
[sudo] password for kali:
(root@kali)-[~]
$ nmap testasp.vulnweb.com/
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 10:45 -0400
Unable to split netmask from target expression: "testasp.vulnweb.com/"
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds

(root@kali)-[~]
$ nmap testasp.vulnweb.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-07 10:45 -0400
Nmap scan report for testasp.vulnweb.com (44.238.29.244)
Host is up (0.019s latency).
Other addresses for testasp.vulnweb.com (not scanned): 64:ff9b::2cee:1df4
rDNS record for 44.238.29.244: ec2-44-238-29-244.us-west-2.compute.amazonaws.com
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 9.52 seconds

(root@kali)-[~]
$
```

Result

Port / Protocol	State	Service	Version
80/tcp	Open	HTTP	nginx 1.19.0

Impact

Advantages

- 1. Users can open the website easily.
- 2. Browsers use port 80 by default.

Disadvantages

- 1. HTTP is not secure.
- 2. Attackers may capture data.
- 3. Open ports can be attacked.

Mitigation

- 1. Use HTTPS instead of HTTP.
- 2. Update the server regularly.
- 3. Monitor network activity.
- 4. Do regular security checks.

Tools Used

1. Kali Linux.
2. Nmap
3. Ping Tool

Task 2 – Directory Brute Forcing

Find hidden folders on the website: **http://testasp.vulnweb.com**

Attack Type : Directory Brute Forcing

Why Is This Important?

- Some folders in a website are hidden. These folders may contain admin pages or important files.
- If attackers find these folders, they may steal data or damage the website.

Step 1 – Choose the Target Website

Target website: **http://testasp.vulnweb.com**

Step 2 – Run Dirbuster

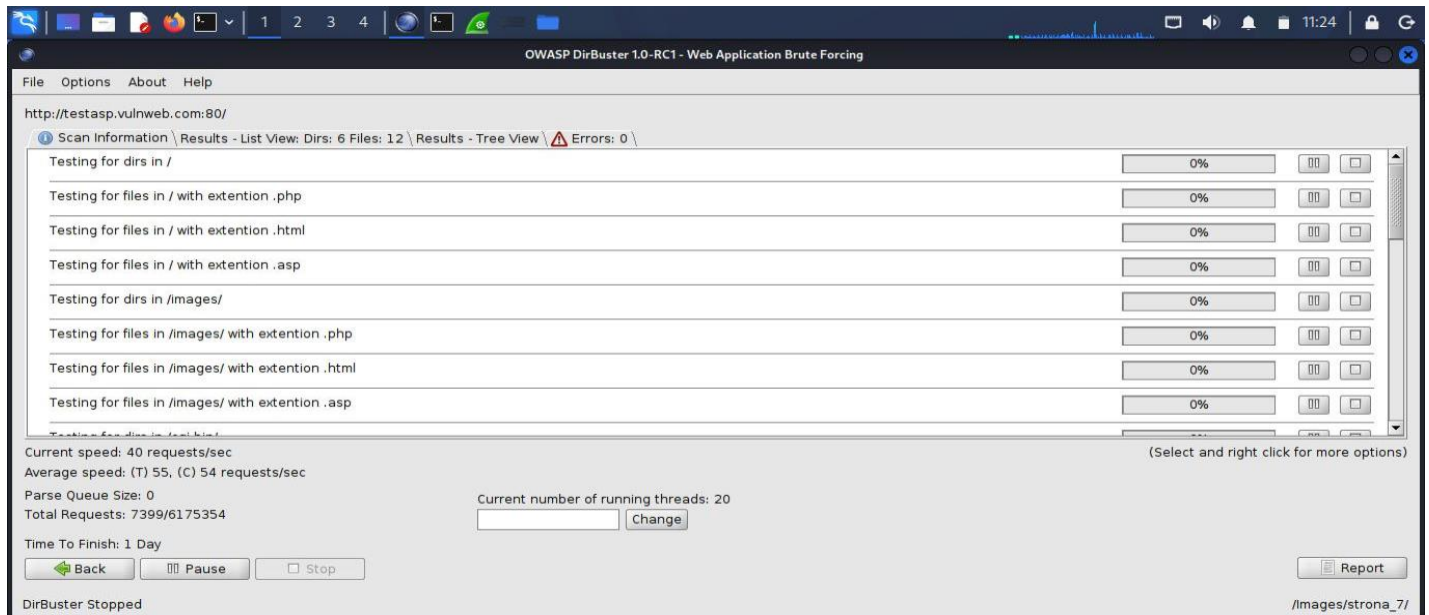
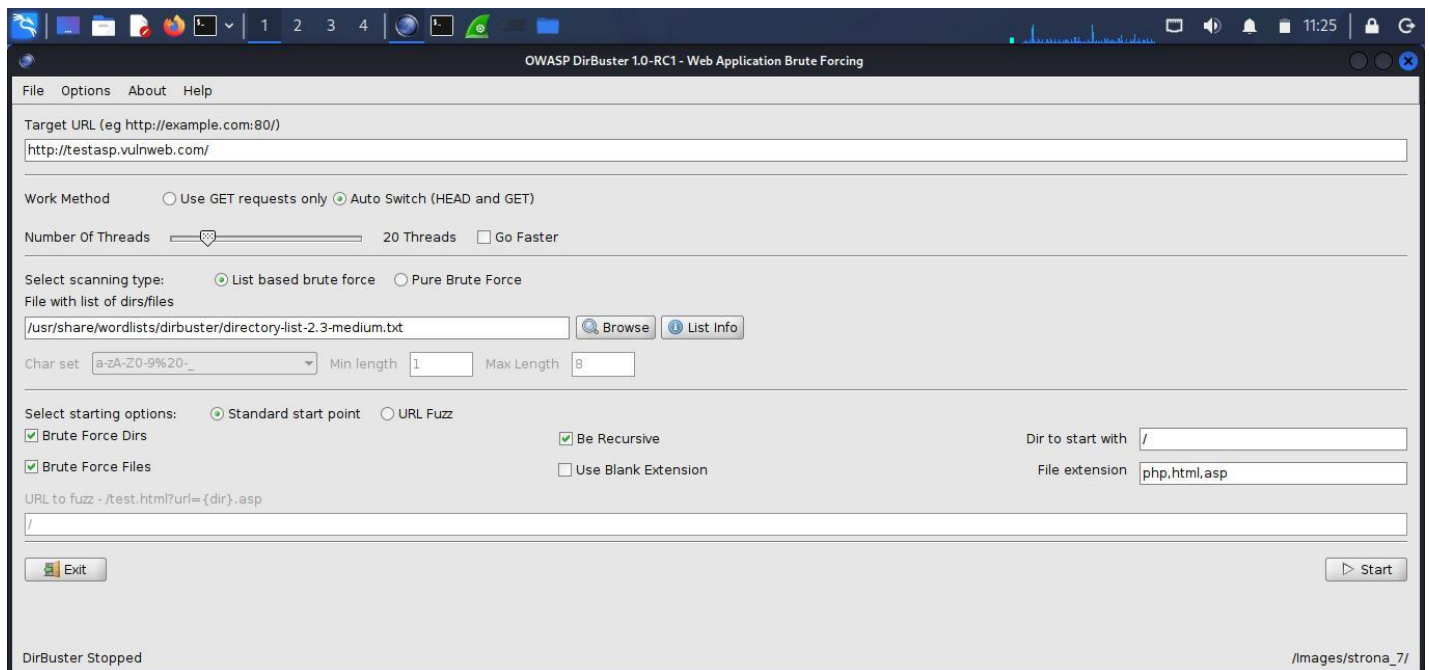
DirBuster was used to find hidden folders.

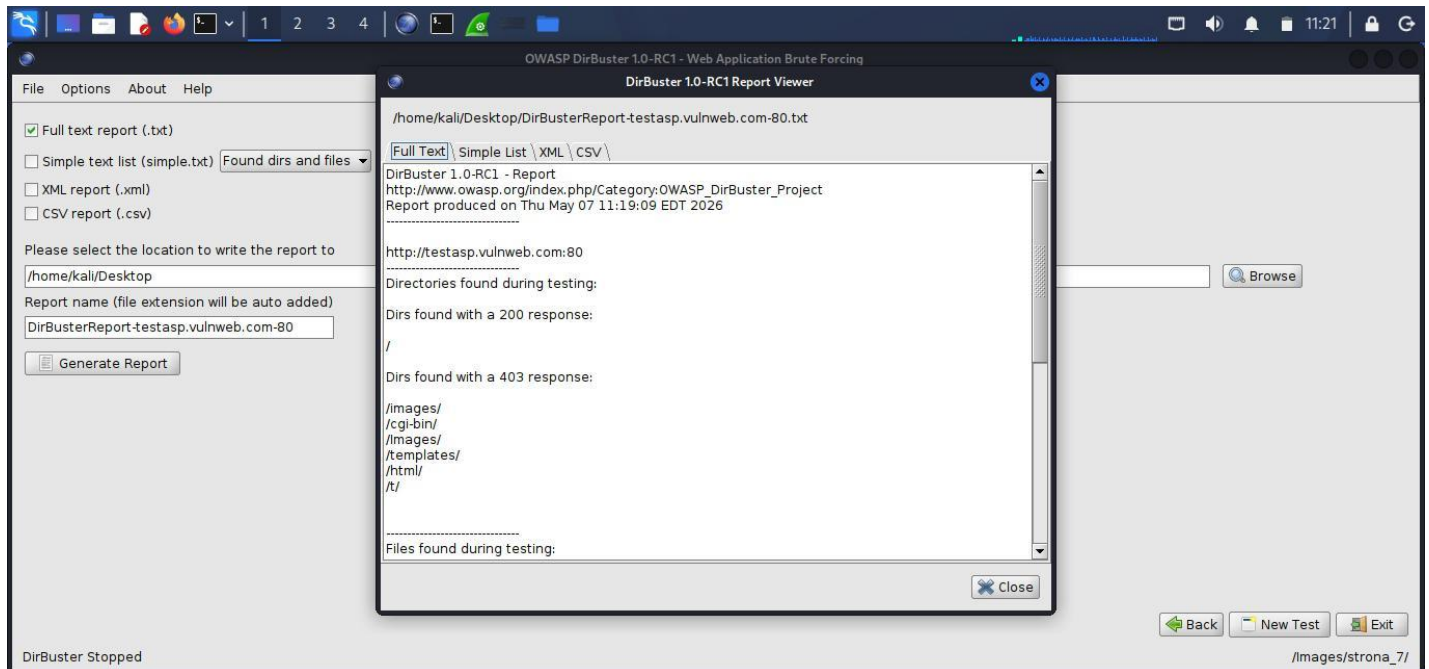
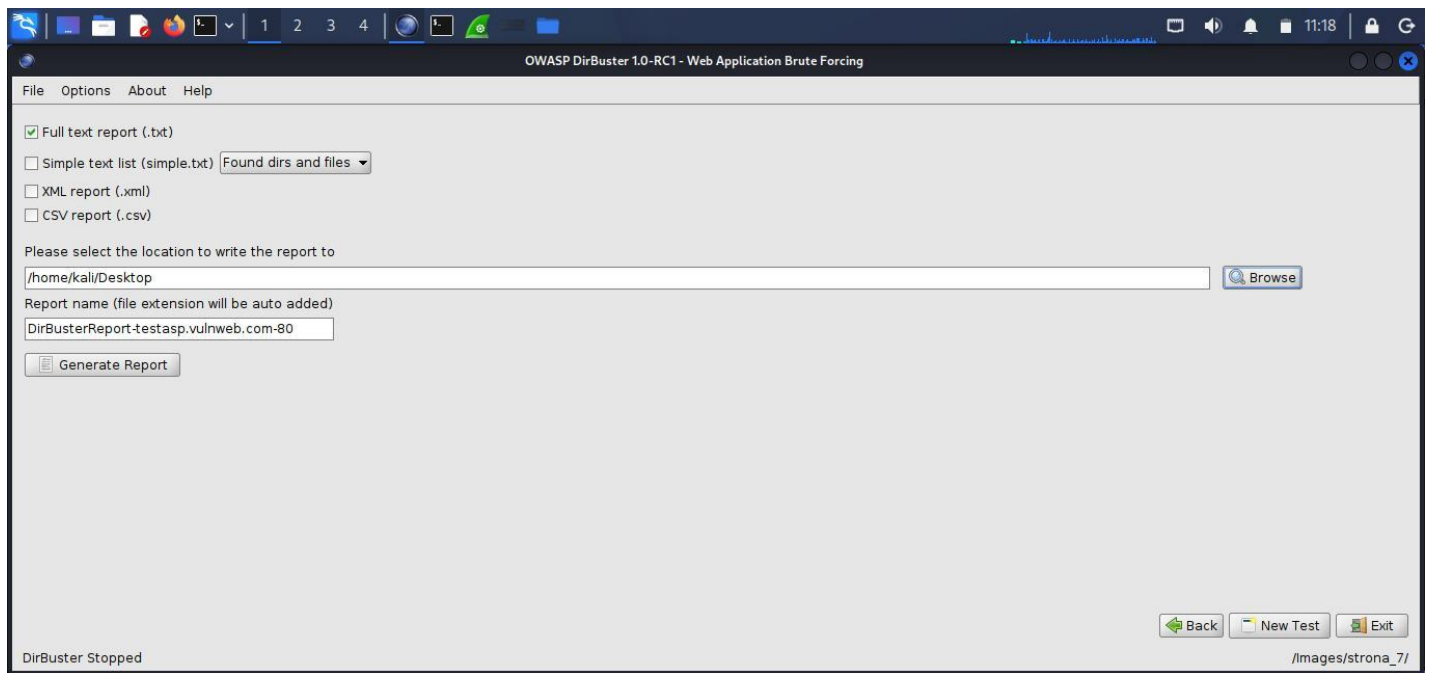
It checks many folder names automatically and finds which folders are available.

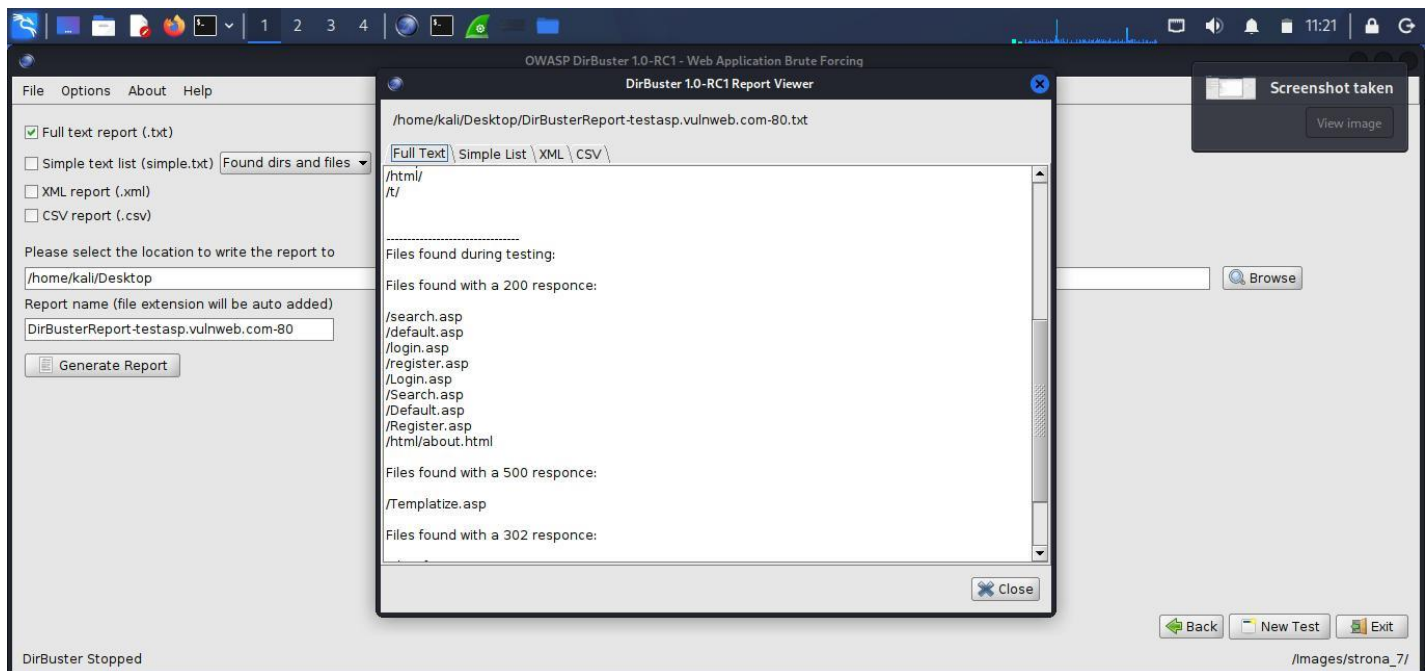
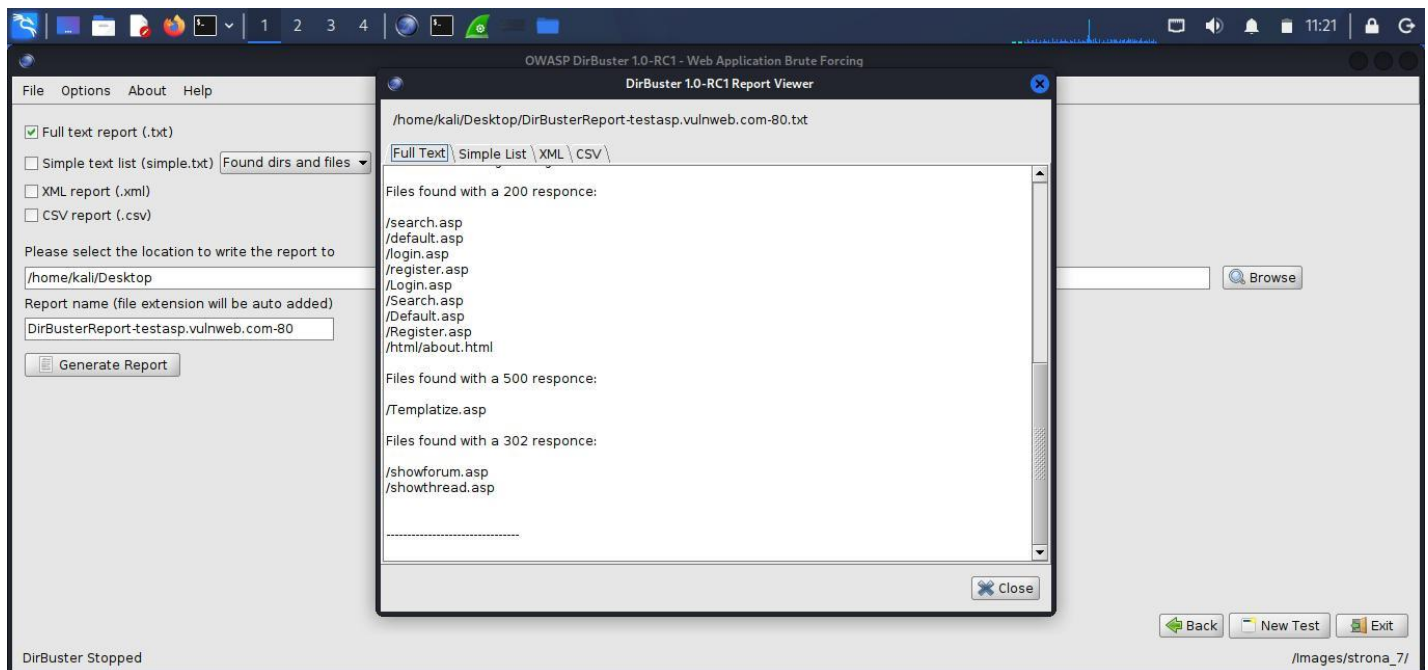
Command Used

Dirb <http://testasp.vulnweb.com>

Step 3 – Dirbuster Scan Results (Screenshots)







Directories Found

- <http://testasp.vulnweb.com/admin/>
- <http://testasp.vulnweb.com/CVS/>
- <http://testasp.vulnweb.com/images/>
- <http://testasp.vulnweb.com/pictures/>

- <http://testasp.vulnweb.com/secured/>
- <http://testasp.vulnweb.com/vendor/>

The most dangerous directories found are /admin, /CVS, /secured, and /vendor, as these may contain critical configuration files or sensitive data.

Impact :

1. Hidden folders may contain important files.
2. Attackers may access private data.
3. The website may be attacked.

Mitigation

1. Protect important folders.
2. Remove unwanted files.
3. Use proper permissions.
4. Monitor website activity regularly.

Tools Used

- Kali Linux
- DirBuster

Task 3 – Network Traffic Sniffing

Capture login details using Wireshark

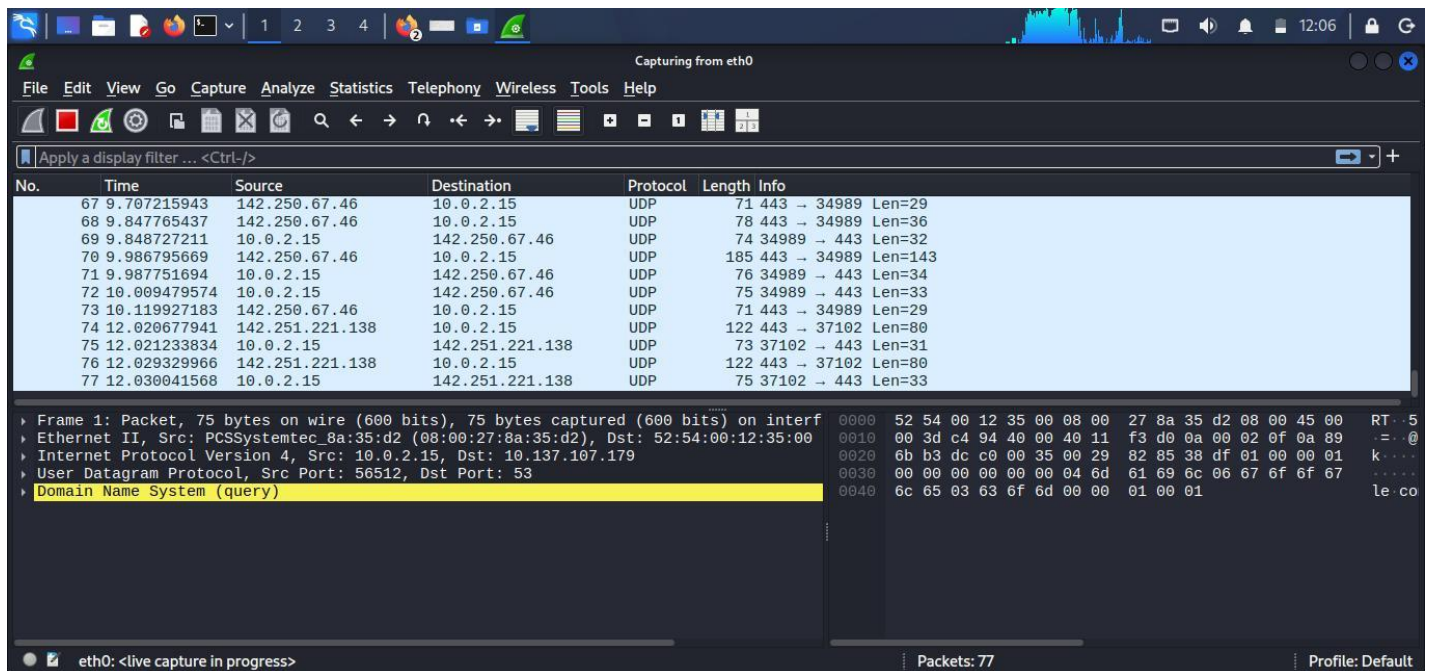
Attack Type : Network Sniffing

Why Is This Important?

- If a website uses HTTP, usernames and passwords are not secure.
- Attackers can capture and read this data from the network.

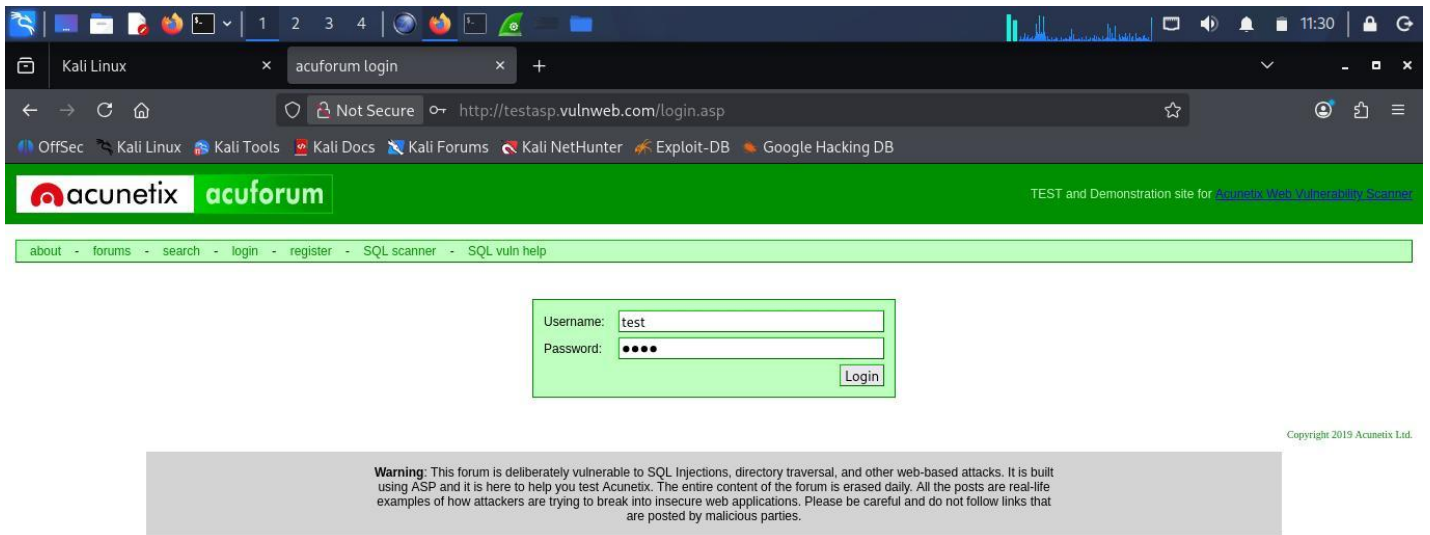
Step 1 – Open Wireshark

- Wireshark was opened in Kali Linux and network capture was started.



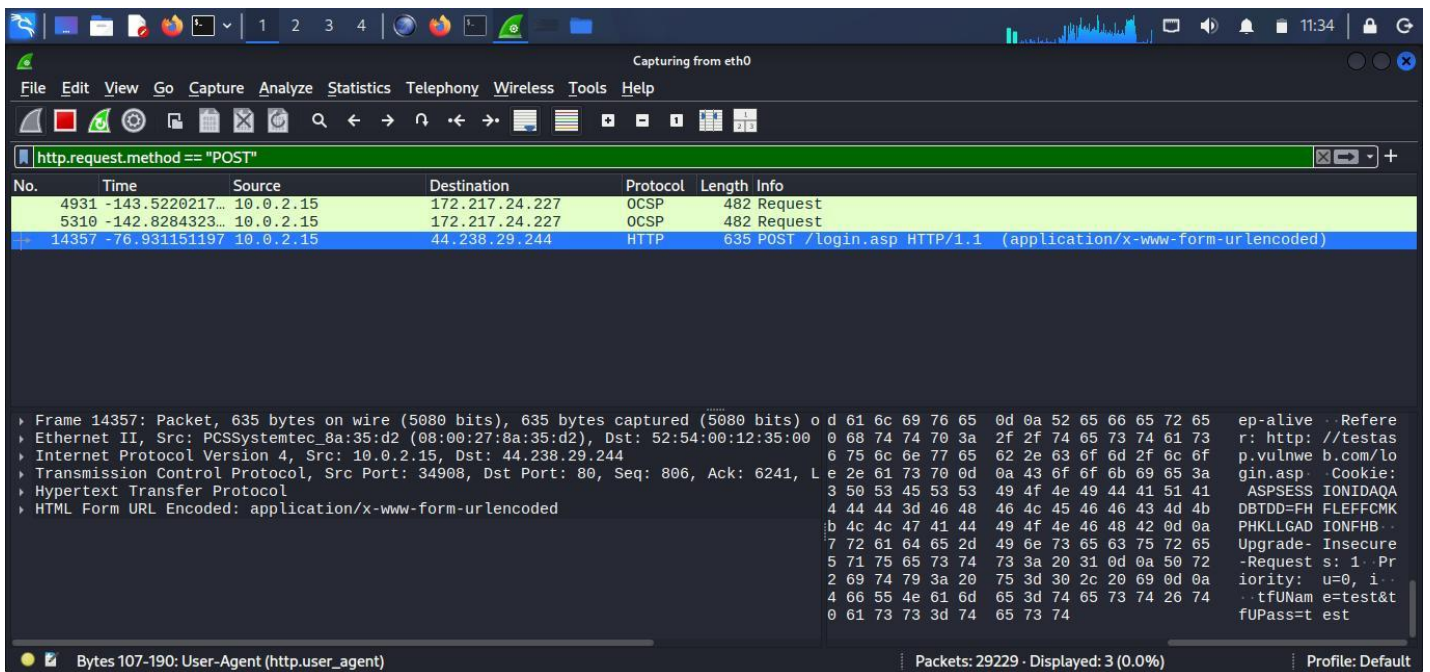
Step 2 – Attempt Login on the Website

- The login page was opened in Firefox:
<http://testasp.vulnweb.com/login.php>
- A username and password were entered.



Step 3 – Check the Captured Packet.

- After login, the capture was stopped.
- The http filter was used to check HTTP traffic.



Step 4 – Find Login Details

- The login packet was found in Wireshark.

- The username and password were visible in plain text.

Captured Credentials

- Username: test
- Password: test

The image shows a Wireshark packet capture of an HTTP login request. The packet list on the left shows Frame 14357, which is an HTTP GET request. The packet details pane on the right shows the request structure, including the User-Agent, Host, and Form parameters. The packet bytes pane at the bottom shows the raw data of the request, which is URL-encoded. The credentials 'test' for both the username and password fields are visible in the raw data.

```

Frame 14357: Packet, 635 bytes on wire (5080 bits), 635 bytes captured (5080 bits) on interface 0
Ethernet II, Src: PCSSystemtec_8a:35:d2 (08:00:27:8a:35:d2), Dst: 52:54:00:12:35:00
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 44.238.29.244
Transmission Control Protocol, Src Port: 34908, Dst Port: 80, Seq: 806, Ack: 6241, Len: 50
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
    ep=live&Referer=http://testasp.vulnweb.com/login.asp&Cookie=ASPSESS=IONIDAQA&DBTDD=FB&FLEFFCMK&PHKLLGAD=IONFHB&Upgrade-Insecure-Request=1&Priority=u=0,i=0&tfUName=test&tfUPass=test
  
```

Impact

1. Attackers may steal login details.
2. User accounts may be hacked.
3. Private data may be stolen.

Mitigation

1. Use HTTPS.
2. Enable MFA.
3. Monitor network traffic.
4. Update software regularly.

Tools Used

- Kali Linux
- Wireshark
- Firefox